1. Introduction
The goal of this project is to develop an AI-driven cold-calling system that automates and streamlines client engagement. The system will leverage NVIDIA Riva for speech-to-text and text-to-speech capabilities, OpenAI for natural language processing, and integrate with Twilio for VoIP. n8n will be used for automation and workflow orchestration. The project aims to provide businesses with an efficient, scalable solution to enhance customer interaction and improve sales automation.

2. Project Objectives
Automate Cold Calling: Reduce manual labor in making initial contact with leads and clients.
AI Integration: Implement OpenAI for conversation generation, ensuring realistic and engaging dialogues.
Voice Interaction: Utilize NVIDIA Riva ASR for accurate speech-to-text and NVIDIA Riva TTS for realistic voice responses.
Automation: Use n8n to automate workflows such as lead scoring, CRM updates, and call logging.
Scalability: Ensure the system can handle multiple calls concurrently and scale as required.
reduce
Latency: Achieve maximum latency < 500

3. Project Scope
3.1. Functional Requirements
Call Handling: The system must be able to initiate and handle inbound/outbound calls.
Speech Recognition: Convert speech to text for understanding client inquiries using NVIDIA Riva ASR.
AI-driven Response: Generate responses using OpenAI's model based on customer input.
Voice Synthesis: Convert generated responses back to speech using NVIDIA Riva TTS.
Workflow Automation: Automate backend processes like updating CRM data and logging call details via n8n.
Data Storage: Efficient storage for call records and client interaction data.
3.2. Non-Functional Requirements
Performance: Achieve a response time (latency) of less than 500ms.
Scalability: The system should scale based on demand and handle high volumes of calls.
Reliability: The system should be robust and capable of handling errors gracefully.
Security: Implement secure data handling practices, especially concerning client information and call recordings.

4. Technical Architecture
The system will be designed using the following architecture:
Frontend: A React/Next.js dashboard for call management, status tracking, and real-time monitoring.
Backend:
FastAPI for API handling.
Node.js for additional processing.
API Gateway for routing requests.
Call Manager for managing the call flow and coordinating with Twilio.
Memory Buffer for storing transcriptions and AI responses.
AI Processing:
NVIDIA Riva ASR for Speech-to-Text.
OpenAI for conversational AI response generation.
NVIDIA Riva TTS for Text-to-Speech synthesis.
Telephony Integration: Use of Twilio for voice service.
Automation: n8n to automate workflows like CRM integration, lead scoring, and call logging.

5. Milestones and Deliverables

Requirement Analysis and Planning	2 weeks	2025-02-01	2025-02-14	Functional & Non-Functional Requirements Document

System Architecture Design	2 weeks	2025-02-15	2025-02-28	Technical Architecture Diagrams, Database Schema
Frontend & Backend Development	6 weeks	2025-03-01	2025-04-15	Functional Frontend and Backend Components
AI Integration & Testing	4 weeks	2025-04-16	2025-05-15	Integrated AI System (ASR, TTS, OpenAI)
VoIP and Automation Integration	3 weeks	2025-05-16	2025-06-05	Working Integration with Twilio and n8n
System Testing & Optimization	3 weeks	2025-06-06	2025-06-26	Full System Testing, Performance Optimization
Deployment and Monitoring Setup	2 weeks	2025-06-27	2025-07-10	Deployment on AWS, Monitoring Dashboard Setup
Project Review & Final Report	1 week	2025-07-11	2025-07-17	Final Project Report, Handover Documentation
Requirement Analysis and Planning	2 weeks	2025-02-01	2025-02-14	Functional & Non-Functional Requirements Document

6. Resource Allocation

6.1. Team Members

Project Manager	Oversee project execution, timeline, and budget	Rafay
Backend Developer	API development, server-side logic, database	Ahsan
Frontend Developer	UI/UX development for dashboard	Arbab, Fakhir
AI Specialist	Integrate NVIDIA Riva and OpenAI	Ahsab, Arbab
VoIP Specialist	Integrate Twilio for call handling	Ahsab, Arbab
Automation Engineer	Set up n8n workflows and CRM integration	Ahsab, Arbab
QA Engineer	Test functionality, performance, and security	Arbab
DevOps Engineer	Deployment, cloud management, scalability	Ahsan

6.2. Budget

AWS EC2 Instance (VPS)	$120/month
Twilio VoIP Integration	$30/month
Deepgram STT TTS	20$
OpenAI API Costs	$50/month

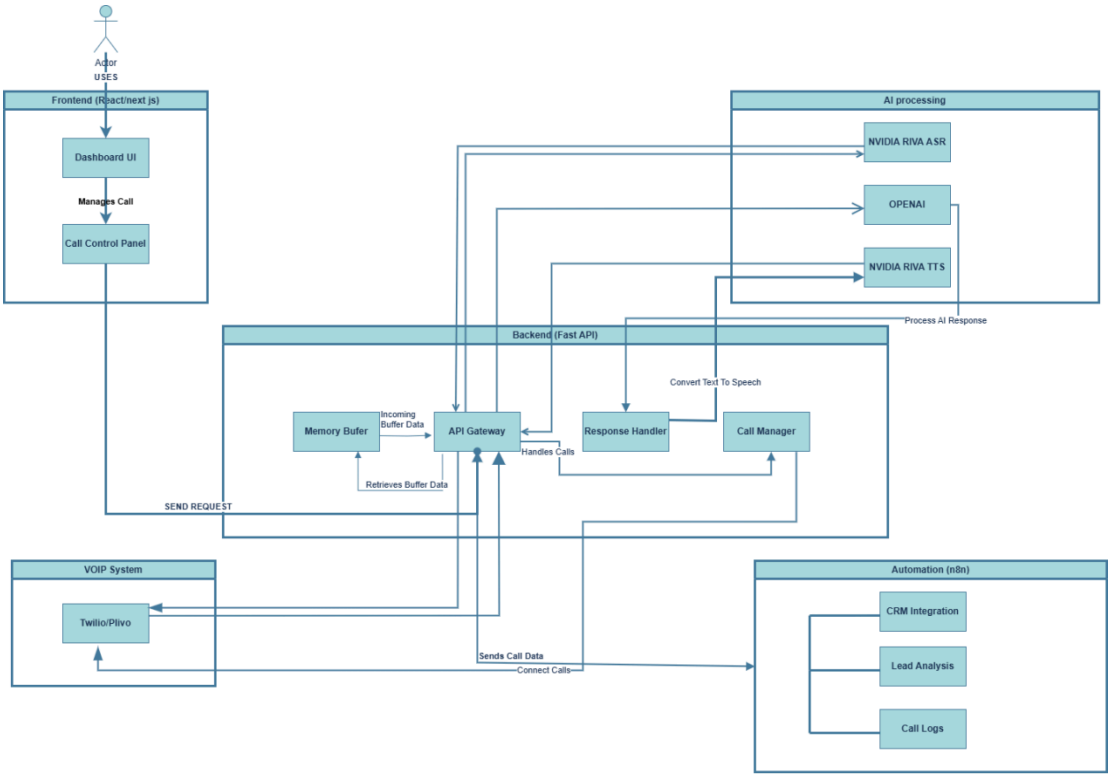
Miscellaneous Costs	$80
Total Operational Project Budget/month	**$300**

7. Risks and Mitigations

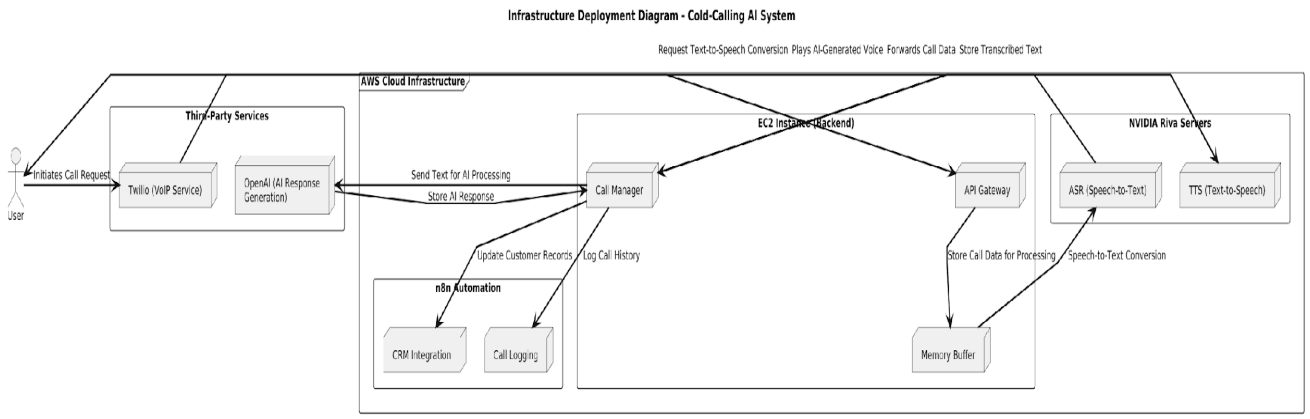
Latency Issues	High	High	Optimize API calls and AI processing flow to achieve <500ms latency.
Integration Challenges	Medium	High	Ensure clear integration protocols for Twilio, NVIDIA Riva, and OpenAI.
Cost Overruns	Medium	Medium	Regular budget reviews, and scale services as needed.
Data Privacy Concerns	High	High	Ensure compliance with GDPR and other privacy regulations.
Deployment Delays	Medium	Medium	Regular progress tracking and timely reviews.

8. Project Diagrams

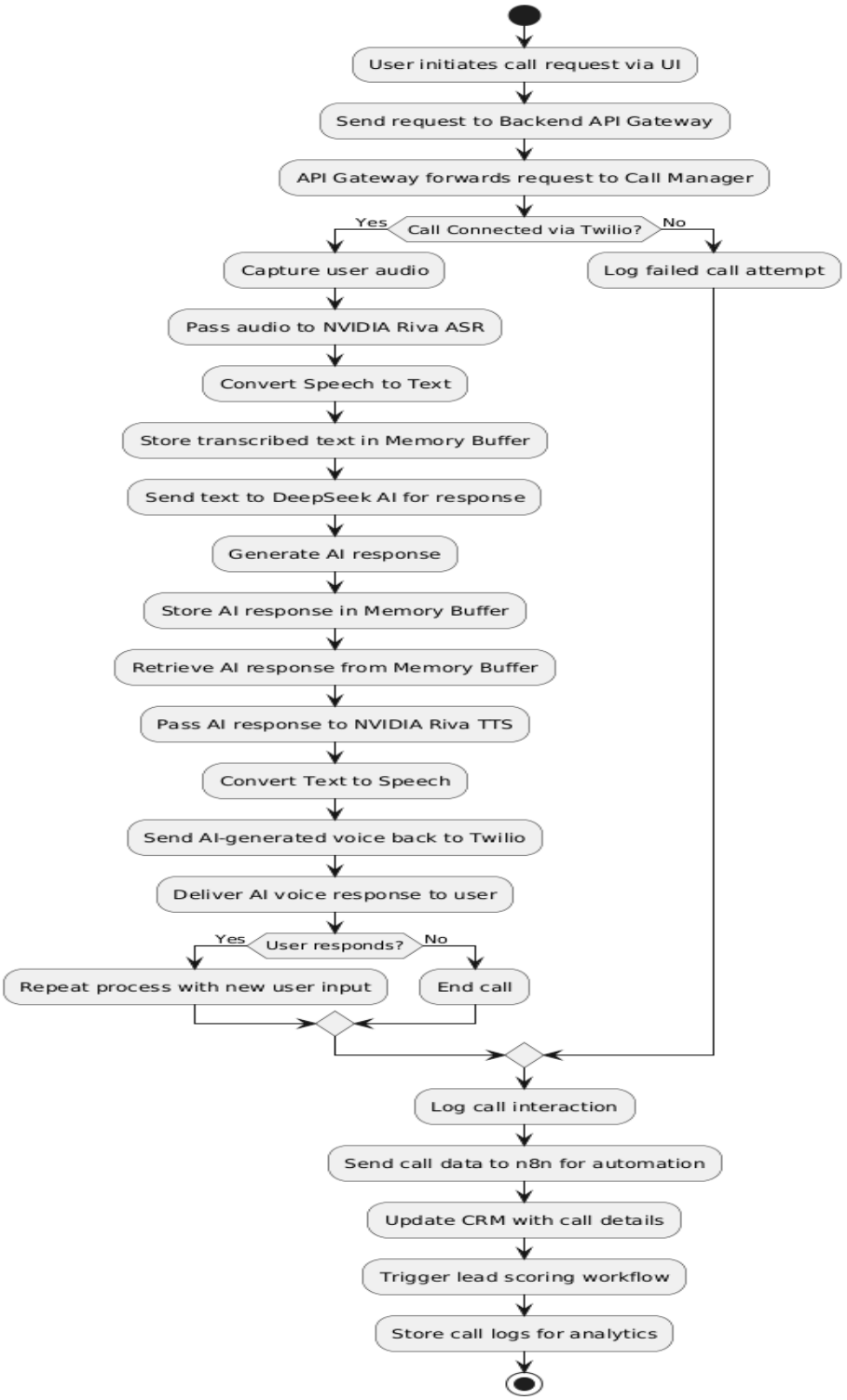
8.1. System Architecture Diagram(UML)



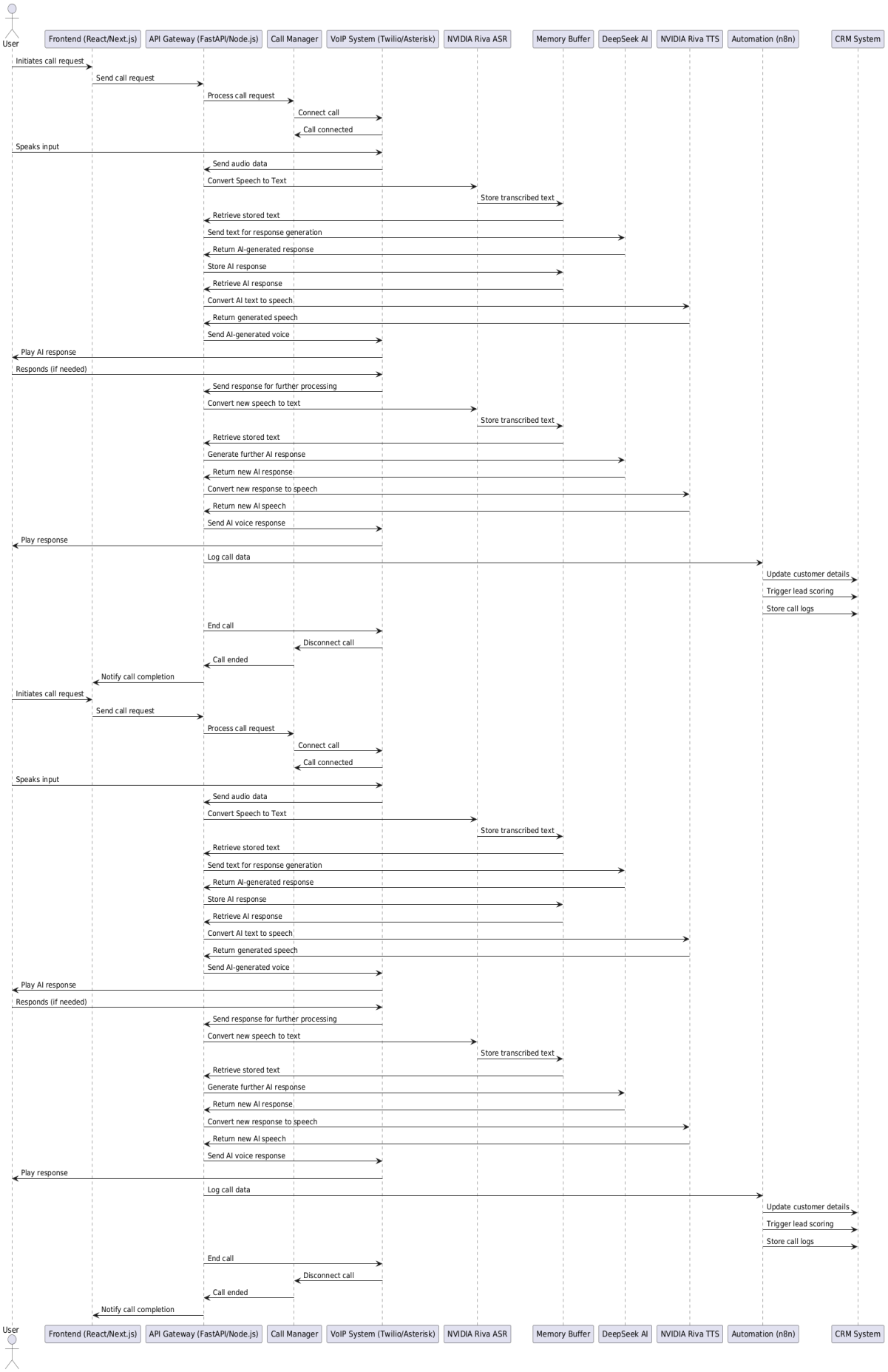
8.2. Infrastructure Deployment diagram



8.3. Activity Diagram



8.4. Sequence Diagram


User
Frontend (React/Next.js)
API Gateway (FastAPI/Node.js)
Call Manager
VoIP System (Twilio/Asterisk)
NVIDIA Riva ASR
Memory Buffer
DeepSeek AI
NVIDIA Riva TTS
Automation (n8n)
CRM System
Initiates call request
Send call request
Process call request
Connect call
Call connected
Speaks input
Send audio data
Convert Speech to Text
Store transcribed text
Retrieve stored text
Send text for response generation
Return AI-generated response
Store AI response
Retrieve AI response
Convert AI text to speech
Return generated speech
Send AI-generated voice
Play AI response
Responds (if needed)
Send response for further processing
Convert new speech to text
Store transcribed text
Retrieve stored text
Generate further AI response
Return new AI response
Convert new response to speech
Return new AI speech
Send AI voice response
Play response
Log call data
Update customer details
Trigger lead scoring
Store call logs
End call
Disconnect call
Call ended
Notify call completion
Initiates call request
Send call request
Process call request
Connect call
Call connected
Speaks input
Send audio data
Convert Speech to Text
Store transcribed text
Retrieve stored text
Send text for response generation
Return AI-generated response
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User
Frontend (React/Next.js)
API Gateway (FastAPI/Node.js)
Call Manager
VoIP System (Twilio/Asterisk)
NVIDIA Riva ASR
Memory Buffer
DeepSeek AI
NVIDIA Riva TTS
Automation (n8n)
CRM System

9. Latency Breakdown:

VoIP Call Setup (Twilio)	10ms
ASR (Speech-to-Text, NVIDIA Riva)	100ms
Memory Buffer Storage & Retrieval	10ms + 10ms
OpenAI Response Generation	150ms
TTS (Text-to-Speech, NVIDIA Riva)	100ms
VoIP Audio Playback	50ms
Automation (n8n, CRM Updates, Logging)	20ms + 30ms

10. Conclusion

This phase of the project focuses on careful planning, ensuring we are equipped with the required resources, understanding the technical architecture, and managing risks effectively. With clear milestones and resource allocation, we aim to successfully implement the cold-calling AI system in a manner that meets both business and technical objectives. The project team is well-prepared for the next steps, and we are confident in achieving the goals within the outlined timeline and budget.